

Categorical Variable Discovery Report

Chi-squared Test of Independence

variable	chi_squared	p_value	cramers_v
payment_type	303.17	< 0.0001	0.039
housing_status	2,285.53	< 0.0001	0.107
employment_status	250.86	< 0.0001	0.035
device_os	1,242.31	< 0.0001	0.079

Each test was performed using the same 100,000 record sample.

The results above suggest there **is** a statistically significant association between each categorical variable and fraud. **However**, most display a weak association with fraud according to Cramers v (< 0.1).

Payment Type Univariate Analysis Results

payment type	log odds	p-value	odds ratio	predicted probability
Intercept (AA)	-5.238	< 0.0001	0.005	0.005
AB	0.762	< 0.0001	2.143	0.682
AC	1.163	< 0.0001	3.200	0.762
AD	0.723	< 0.0001	2.061	0.673
AE	-0.425	0.6720	0.654	0.395

Note: Payment type AA is the reference category for the log odds & odds ratios.

The final ordering of payment categories from **most to least** associated with fraud is:
[AC, AB, AD, AE, AA].

Housing Status Univariate Analysis Results

housing status	log odds	p-value	odds ratio	predicted probability
Intercept (BA)	-3.246	< 0.0001	0.039	0.038
BB	-1.862	< 0.0001	0.155	0.134
BC	-1.839	< 0.0001	0.159	0.137
BD	-1.497	< 0.0001	0.224	0.183
BE	-2.422	< 0.0001	0.089	0.082
BF	-2.224	< 0.0001	0.108	0.097
BG	-2.279	0.0230	0.102	0.093

Note: Housing status BA is the reference category for the log odds & odds ratios.

The final ordering of housing categories from **most to least** associated with fraud is:

[BA, BD, BC, BB, BF, BG, BE].

Employment Status Univariate Analysis Results

employment status	log odds	p-value	odds ratio	predicted probability
Intercept (CA)	-4.395	< 0.0001	0.012	0.012
CB	-0.575	< 0.0001	0.563	0.360
CC	0.719	< 0.0001	2.052	0.672
CD	-1.182	< 0.0001	0.307	0.235
CE	-1.662	< 0.0001	0.190	0.160
CF	-1.853	< 0.0001	0.157	0.136
CG	0.241	0.5270	1.273	0.560

Note: Employment status CA is the reference category for the log odds & odds ratios.

The final ordering of employment categories from **most to least** associated with fraud is:

[CC, CG, CA, CB, CD, CF, CE].

Device Operating System Univariate Analysis Results

device os	log odds	p-value	odds ratio	predicted probability
Intercept (Linux)	-5.263	< 0.0001	0.005	0.005
macintosh	1.006	< 0.0001	2.735	0.732
other	0.112	0.0010	1.119	0.528
windows	1.587	< 0.0001	4.889	0.830
x11	0.783	< 0.0001	2.188	0.686

Note: Device OS Linux is the reference category for the log odds & odds ratios.

The final ordering of device os categories from **most to least** associated with fraud is:
[Windows, Macintosh, X11, Other, Linux].